

Example 1

$S \rightarrow aAb$

$aA \rightarrow aaAb$

$A \rightarrow \varepsilon$

- Generated language: $\{a^n b^n \mid n > 0\}$
- OBSERVE: Different grammars can generate the same language

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Example 2

$S \rightarrow AB$

$A \rightarrow aA$

$A \rightarrow a$

$B \rightarrow Bb$

$B \rightarrow b$

- Generated language: $\{a^n b^m \mid n, m > 0\}$

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Example 3

$S \rightarrow aSBc$

$S \rightarrow abc$

$cB \rightarrow Bc$

$bB \rightarrow bb$

- Generated language: $\{a^n b^n c^n \mid n > 0\}$

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Example 4

$S \rightarrow AB$

$A \rightarrow a$

- Generated language: $\emptyset$

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Example 5

$$S \rightarrow \varepsilon$$

- Generated language: $\{\varepsilon\}$
- $\{\varepsilon\} \neq \emptyset$

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Example 6

$$S \rightarrow aSb$$

$$S \rightarrow \varepsilon$$

- Generated language:
 - $\{a^n b^n \mid n > 0\} \cup \{\varepsilon\} = \{a^n b^n \mid n \geq 0\}$

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Example

Notation for more productions
with same left-hand side

$S \rightarrow CD$
 $C \rightarrow aCA \mid bCB$
 $AD \rightarrow aD$
 $BD \rightarrow bD$
 $Aa \rightarrow aA$
 $Ab \rightarrow bA$
 $Ba \rightarrow aB$
 $Bb \rightarrow bB$
 $C \rightarrow \varepsilon$
 $D \rightarrow \varepsilon$

- Generated language?

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Example 7: Derivation 1

$S \rightarrow CD$
 $C \rightarrow aCA \mid bCB$
 $AD \rightarrow aD$
 $BD \rightarrow bD$
 $Aa \rightarrow aA$
 $Ab \rightarrow bA$
 $Ba \rightarrow aB$
 $Bb \rightarrow bB$
 $C \rightarrow \varepsilon$
 $D \rightarrow \varepsilon$

$S \Rightarrow CD$
 $CD \Rightarrow D$
 $D \Rightarrow \varepsilon$

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Example 7: Derivation 2

$S \rightarrow CD$

$C \rightarrow aCA \mid bCB$

$AD \rightarrow aD$

$BD \rightarrow bD$

$Aa \rightarrow aA$

$Ab \rightarrow bA$

$Ba \rightarrow aB$

$Bb \rightarrow bB$

$C \rightarrow \varepsilon$

$D \rightarrow \varepsilon$

$S \Rightarrow CD$

$CD \Rightarrow aCAD$

$aCAD \Rightarrow aCaD$

$aCaD \Rightarrow aaD$

$aaD \Rightarrow aa$

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Example 7: Derivation 3

$S \rightarrow CD$

$C \rightarrow aCA \mid bCB$

$AD \rightarrow aD$

$BD \rightarrow bD$

$Aa \rightarrow aA$

$Ab \rightarrow bA$

$Ba \rightarrow aB$

$Bb \rightarrow bB$

$C \rightarrow \varepsilon$

$D \rightarrow \varepsilon$

$S \Rightarrow CD$

$CD \Rightarrow aCAD$

$aCAD \Rightarrow abCBAD$

$abCBAD \Rightarrow abCBA$

$abCBA \Rightarrow abBA$

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Example 7: Derivation 4

$S \rightarrow CD$

$C \rightarrow aCA \mid bCB$

$AD \rightarrow aD$

$BD \rightarrow bD$

$Aa \rightarrow aA$

$Ab \rightarrow bA$

$Ba \rightarrow aB$

$Bb \rightarrow bB$

$C \rightarrow \varepsilon$

$D \rightarrow \varepsilon$

$S \Rightarrow CD$

$CD \Rightarrow aCAD$

$aCAD \Rightarrow abCBAD$

$abCBAD \Rightarrow abCBaD$

$abCBaD \Rightarrow abCaBD$

$abCaBD \Rightarrow abCabD$

$abCabD \Rightarrow ababD$

$ababD \Rightarrow abab$

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Generative Grammars Formally

- A grammar is a tuple
 (V, T, S, P)

- V vocabulary of terminals and non-terminals
- T set of terminals
- S start symbol in $(V \setminus T)$
- P set of productions

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Not

Zero or more repetitions of elements in the base set

- Uppercase, early in the alphabet
 - $A, B, \dots \in (V \setminus T)$
- Uppercase, late in the alphabet
 - $X, Y, \dots \in V$
- Lowercase, early in the alphabet
 - $a, b, \dots \in T$
- Lowercase, early in Greek alphabet
 - $\alpha, \beta, \dots \in V^*$
- Strings of terminals
 - $w, w_0, \dots$

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Productions

- General form:

$$\delta \rightarrow \beta$$

One or more repetitions of elements in the base

- $\delta \in V^+$
- δ contains at least a non-terminal
- δ called **driver** of the production
- β called **body** of the production

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Generated Languages

- $G = (V, T, S, P)$

- $L(G) = \{ w \mid w \in T^* \text{ and } S \Rightarrow^* w \}$

T^* because w may just be ϵ

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Hierarchy of Grammars

- Depending on the shape of productions

- **Context-free grammars**, or just **free grammars**:

- Every production has the form
- $A \rightarrow \beta$
- namely: "non-terminal" arrow "either ϵ or string of terminals and non-terminals"

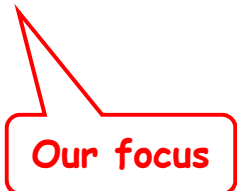
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Context-free Languages

- L is a context-free language
- Iff
- There exists a context-free grammar G such that $L=L(G)$

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Context-free Languages



Our focus

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Canonical Derivations

- **Rightmost (Leftmost) derivation** step:
 - Replace the rightmost (**leftmost**) non-terminal
- **Canonical derivations** of words in the language:
 - Either every derivation step is rightmost
 - Or every derivation step is leftmost

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Derivation Trees

- Start symbol is the root
- For every derivation step under the production
- $A \rightarrow X_1 X_2 \dots X_n$
- Generate children $X_1 X_2 \dots X_n$ for node A
- Terminals are the leaves (and so is ε)

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